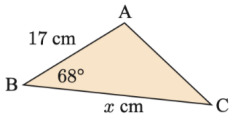
EXERCISE 1

[3 marks]

If triangle ABC has area 150 cm^2, find the value of x.



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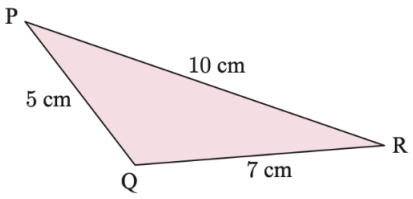
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EXERCISE 2

[5 marks]

a. Find the measure of obtuse $\widehat{PQR}$.

b. Hence find the area of $\triangle PQR$.



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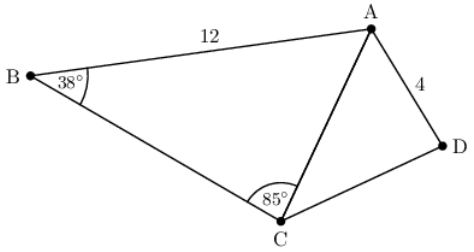
EXERCISE 3 [4 marks]

Triangle ABC has angle $B = 40°$, $b = 8$ cm, and $c = 11$ cm. Find the two possible measures of angle C.

EXERCISE 4

[8 marks]

The following diagram shows quadrilateral ABCD.



$\mathrm{AB} = 12\ \mathrm{cm},\ \mathrm{AD} = 4\ \mathrm{cm},\ \hat{\mathrm{ACB}} = 85^\circ,\ \hat{\mathrm{CBA}} = 38^\circ.$

1. Find AC.
2. Find the area of triangle ABC.

The area of triangle ABC is three times bigger than the area of triangle ACD.

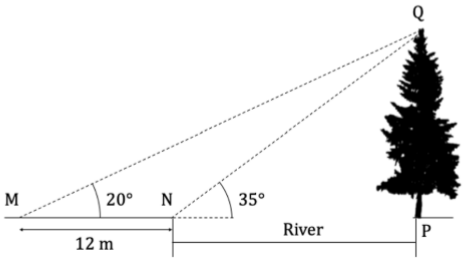
1. Find the acute angle $\hat{\mathrm{CAD}}$.
2. Find CD.

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EXERCISE 5

1. [3 marks]

Adah would like to estimate the height of a tree located at point P on the edge of a riverbank, with the top of the tree at point Q. However, due to a raging river, she is unable to reach the base of the tree. From point M she measures an angle of elevation of 20° to the top of the tree, and then from point N (which is on the edge of Adah's bank of the river) she measures an angle of elevation of 35° to the top of the tree. Between the points M and N she measures a horizontal distance of 12 m. Points M, N and P all lie on a single horizontal line, and point Q is vertically above point P. The diagram below shows this information.



Calculate the length of NQ.

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2. [2 marks]

Calculate the height of the tree.

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3. [3 marks]

Adah borrows a boat and crosses the river at a rate of 50 metres per 15 minutes.

Assuming that she crosses in a straight line directly from point N to point P, find out how long it takes her to cross the river.

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